

src/app/mermaidBoot.ts

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Overview

`src/app/mermaidBoot.ts` (lines 1-37) introduces a lazy-initialisation helper for Mermaid.

It detects the browser's color-scheme preference, caches the chosen theme, and exposes a public API to render Mermaid diagrams on a list of DOM nodes.

Key changes

- `initTheme` – module-level cache (`"dark" | "default" | null`) to remember the last used theme (R1-3).
- `currentMermaidTheme()` – checks `window.matchMedia("(prefers-color-scheme: dark)")`; returns `"default"` when the user prefers light mode or when `window` is undefined (SSR) (R3-11).
- `ensureMermaidInit()` – asynchronously imports `mermaid`, calls `m.initialize` with `startOnLoad: false`, the detected theme, and other options, then updates `initTheme`. It exits early if the theme hasn't changed (R14-27).
- `runMermaidNodes(nodes)` – public API that first ensures initialization, then imports `mermaid` again (to satisfy type-checking) and runs `m.run({ nodes })`. It is a no-op for an empty array (R30-37).

Impact

- **Correctness** – caching prevents duplicate initialisation per theme.
- **Performance** – lazy imports and early return on unchanged theme reduce startup cost.
- **Dark-mode support** – `currentMermaidTheme` automatically selects `"default"` or `"dark"` based on system preference.
- **SSR safety** – checks for `window` existence, so the module can be imported in server-side contexts without throwing.
- **Observability** – no global side effects; initialisation is explicit via `ensureMermaidInit`.

Risks & follow-ups

- **Race conditions** – concurrent calls to `ensureMermaidInit` could trigger multiple imports; verify that the caching logic is sufficient in high-concurrency scenarios.
- **Missing `mermaid` dependency** – dynamic import may fail if the package is not bundled; run a build test to confirm the import path resolves.
- **Empty node handling** – `runMermaidNodes` silently returns for an empty array; ensure callers don't rely on a promise resolution that indicates work was done.
- **Theme mismatch** – if the user toggles the system theme after initialisation, Mermaid will not update automatically; consider adding a listener if dynamic updates are required.